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Reinforced Galton-Watson processes: Malthusian growth, survival and distribution of the population

Bastien Mallein
joint works with Jean Bertoin

IMT – Université de Toulouse
bastien.mallein@math.univ-toulouse.fr

Discrete Random Structures II



1 Malthusian growth, survival and distribution of a (regular) Galton-Watson process

2 The reinforced Galton-Watson process

3 Statement of the results

- Growth rate of the mean population size
- Survival of the process
- Ancestral distribution of the population



Let ν be a probability distribution on $\mathbb{Z}_+$.

Definition

A ν -Galton-Watson tree (or $\text{GW}(\nu)$) is a population model in which each individual, independently from every other, gives birth to a random number of children distributed according to the law ν . The tree starts from an initial individual called the root.

Notation

For all $n \in \mathbb{N}$, we write Z_n the number of individuals alive at generation n . The process $(Z_n, n \geq 1)$ is a Markov process called the ν -Galton-Watson process.



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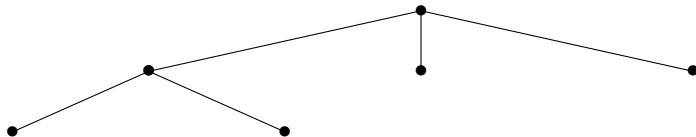


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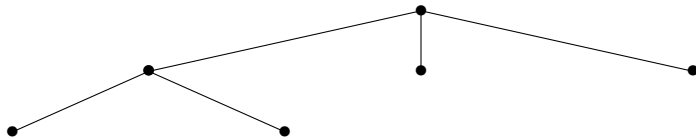


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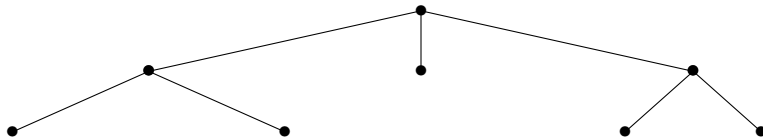


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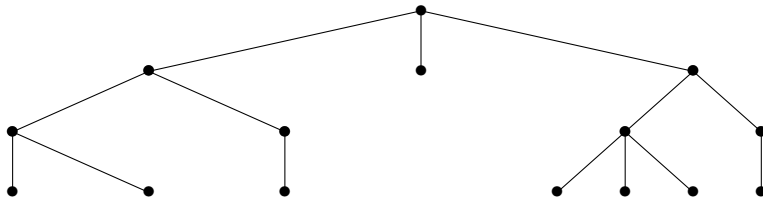


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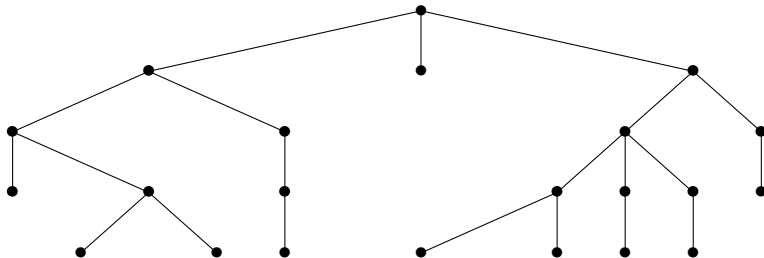


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Mean number of children

We write $m = E(Z_1) = \sum_{j=0}^n j\nu(j)$.

Theorem (Bienaymé 1845, Galton–Watson 1870)

We have $E(Z_n) = m^n$. Moreover,

$$P(\forall n \in \mathbb{N}, Z_n > 0) > 0 \iff m > 1 \quad \text{or} \quad \nu = \delta_1.$$

Proposition

The martingale (Z_n/m^n) converges a.s. to a non-negative limit W . Moreover,

$$P(W > 0) = P(\forall n \in \mathbb{N}, Z_n > 0).$$



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Definition

For u an individual of a tree T , we denote by $L_u = \frac{1}{|u|} \sum_{i=0}^{|u|-1} \delta_{N(u_i)}$ the empirical distribution of the number of children along the ancestral line of this individual.

This encodes the way the ancestors of u reproduced.

Definition

Let T be a (deterministic or random) tree. A probability distribution ρ is called

- ① *evanescent* if there exists a neighbourhood G of ρ such that $\#\{u \in T : L_u \in G\} < \infty$;
- ② *weakly persistent* if for all neighbourhood G of ρ , we have $\#\{u \in T : L_u \in G\} = \infty$;
- ③ *strongly persistent* if there exists an infinite spine (v_n) in T such that $\lim_{n \rightarrow \infty} L_{v_n} = \rho$.



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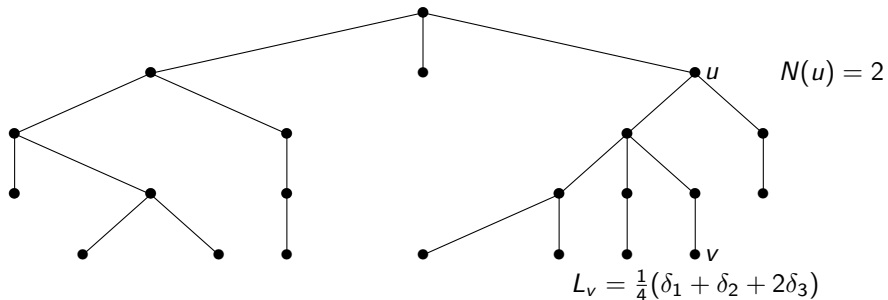


Figure: Notation for Galton-Watson tree

Ancestral distribution of the population

Concentration of the population



We write $\langle \rho, f \rangle = \sum_{k \geq 0} \rho(k) f(k)$. In particular, $\langle \rho, \ln \rangle = \sum_{k \geq 0} \rho(k) \ln(k)$ is $-\infty$ if $\rho(0) > 0$, non-negative otherwise. We also write $H(\mu|\rho) = \sum \mu(j) \log \frac{\mu(j)}{\rho(j)}$.

Theorem (... , Azaïs–Henry ('25), Bertoin–M. ('25+))

Let ν be a probability measure, that we assume to have finite support. Set $\bar{\nu}(k) = \frac{k\nu(k)}{m}$ the size-biased distribution of ν . Let T be a ν -GW. We have:

- for all neighbourhood G of $\bar{\nu}$, there exists $\varepsilon > 0$ such that

$$\mathbb{E}(\#\{|u| = n : L_u \notin G\}) \leq e^{-\varepsilon n} \mathbb{E}(Z_n);$$

- a law ρ is evanescent almost surely if $\langle \rho, \ln \rangle < H(\rho|\bar{\nu})$;
- a law ρ is strongly persistent with positive probability if $\langle \rho, \ln \rangle \geq H(\rho|\bar{\nu})$.

An important lemma

The many-to-one lemma



Lemma

Let T be a (deterministic) rooted tree, we write $h = (h_0, h_1, \dots)$ an *harmonic line* of descent so that $h_0 = \emptyset$ and for all $k \geq 0$, h_{k+1} is a uniformly sampled child of h_k . Let $(x_0, \dots, x_{n-1}) \in \mathbb{N}^n$, setting $\mu = \frac{1}{n} \sum_{j=0}^{n-1} \delta_{x_j}$ we have

$$\#\{|v| = n : N(v_j) = x_j, 0 \leq j \leq n-1\} = P(N(h_j) = x_j, 0 \leq j \leq n-1) \exp(n \langle \mu, \ln \rangle).$$

Corollary

For a $GW(\nu)$ tree T , we have

$$E(\#\{|v| = n : L_v = \mu\}) = P(\frac{1}{n} \sum_{j=0}^{n-1} \delta_{X_j} = \mu) \exp(n \langle \mu, \ln \rangle)$$

where (X_j) are i.i.d. random variables with law ν .

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As a summary:

- The GW process can survive with positive probability iff $m > 1$.
- We have $E(Z_n) \sim_{n \rightarrow \infty} m^n$.
- We have $Z_n = WE(Z_n)(1 + o(1))$ a.s. as $n \rightarrow \infty$.
- Most individuals have an ancestral lineage close to $\bar{\nu}$.
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- with probability $1 - q$, they give birth to a random number of children distributed according to the law ν .
- with probability q , they create the same number of children as one of their ancestors, selected uniformly at random.



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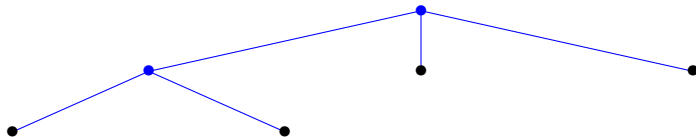


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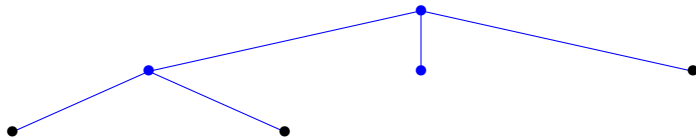


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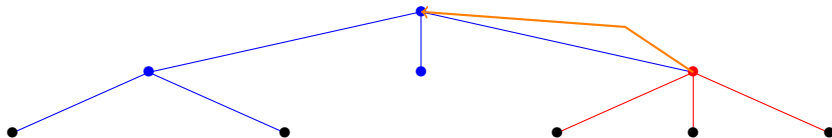


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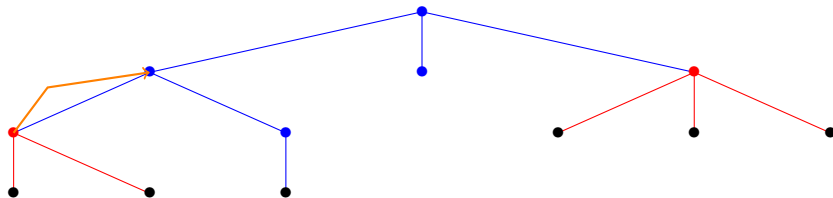


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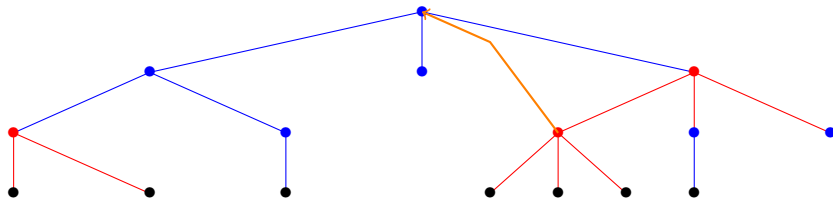


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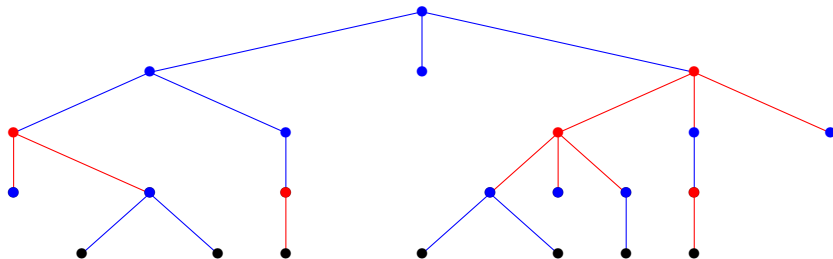


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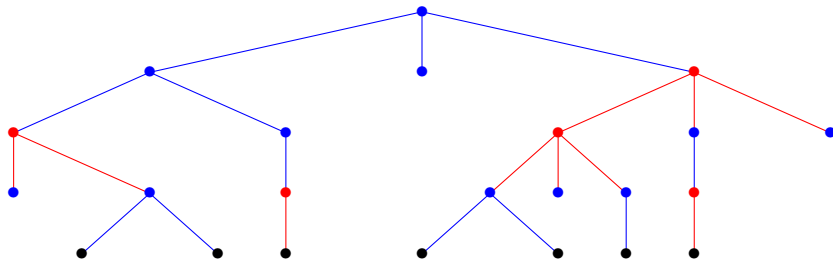


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Reinforced Galton-Watson process



Preliminary remarks

- Natural model for inheritance of fitness traits.
- A $\text{rGW}(\nu, 0)$ is a $\text{GW}(\nu)$. A $\text{rGW}(\nu, 1)$ is with probability $\nu(k)$ a k -ary tree.
- An individual reproducing the offspring of an ancestor always make at least one child.
- All the usual properties of GW trees, such as branching, or the Markov property of (Z_n) , are lost.

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Regular tree

Write $\nu = p\delta_k + (1 - p)\delta_0$. Children in the $\text{rGW}(\nu, q)$ have k children with probability $(1 - q)p + q$.

$$P_q(\forall n \in \mathbb{N}, Z_n > 0) > 0 \iff ((1 - q)p + q)k > 1.$$

- In this situation, the $\text{rGW}(q, \nu)$ can survive even if the $\text{GW}(\nu)$ does not.
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Large support

Assume that there exists $k \in \mathbb{Z}_+$ such that $\nu(k) > 0$ and $((1 - q)\nu(k) + q)k > 1$. Then the subtree consisting of individuals with exactly k children survives with positive probability.

$$\exists k \in \mathbb{Z}_+ : \nu(k) > 0 \text{ and } ((1 - q)\nu(k) + q)k > 1 \Rightarrow P(\forall n \in \mathbb{N}, Z_n > 0) > 0.$$

In particular, if $q > 0$ and ν has unbounded support, then the reinforced Galton-Watson process survives with positive probability.

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From now on, we always assume that ν has finite support, and we denote by k_* the largest integer such that $\nu(k_*) > 0$.



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From now on, we always assume that ν has finite support, and we denote by k_* the largest integer such that $\nu(k_*) > 0$.



- 1 What is the asymptotic growth rate of $E_q(Z_n)$, i.e. $\lim_{n \rightarrow \infty} E_q(Z_n)^{1/n}$?
- 2 Under which conditions on (ν, q) does (Z_n) survives with positive probability?
- 3 What is the a.s. growth rate of (Z_n) , conditionally on survival?
- 4 What is the typical ancestral line of an individual at a large generation?

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For $t < 1/k_*$, set $\Pi(t) = \prod_{j=1}^{k_*} (1 - tj)^{\frac{(1-q)\nu(j)}{q}}$ and $m_{\nu,q} := \frac{q}{\int_0^{1/k_*} \Pi(t) dt}$.

Theorem (Bertoin–M. '24)

For all $q \in (0, 1)$ and ν probability measure on $\mathbb{Z}_+$ with finite support, there exists $m_{\nu,q} > 0$ such that

$$\lim_{n \rightarrow \infty} \mathbb{E}_q(Z_n)^{1/n} = m_{\nu,q}.$$

Much more precisely, we have $\mathbb{E}_q(Z_n) \sim \frac{\nu(k_*)}{q + \nu(k_*)(1-q)} m_{\nu,q}^n$ as $n \rightarrow \infty$.

Growth rate of the mean of a reinforced Galton-Watson process



Some example values

Average growth rate

The formula for $m_{\nu,q}$ appears intricate and is quite difficult to compute in general. However, for some examples these can be computed explicitly.

- If $\nu = (\delta_1 + \delta_2)/2$ and $q = 1/3$, then $m_{\nu,q} = \frac{8}{5}$.
- If $\nu = (\delta_1 + \delta_2)/2$ and $q = 1/5$, then $m_{\nu,q} = \frac{48}{31}$.
- If $\nu = (\delta_0 + \delta_1 + \delta_2 + \delta_3)/4$ and $q = 1/5$, then $m_{\nu,q} = \frac{162}{95}$.

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- If $\nu = (\delta_1 + \delta_2)/2$ and $q = 1/3$, then $m_{\nu,q} = \frac{8}{5}$.
- If $\nu = (\delta_1 + \delta_2)/2$ and $q = 1/5$, then $m_{\nu,q} = \frac{48}{31}$.
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Growth rate of the mean of a reinforced Galton-Watson process



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- 1 Use many-to-one lemma to compute $E(Z_n) = E(\prod_{j=0}^n A_j)$ where (A_n) is a random reinforced sequence.
- 2 Use continuous-time embedding to obtain an ODE system satisfied by $F(t) = e^{-t} \sum_{n \geq 1} (1 - e^{-t})^{n-1} E(Z_n)$.
- 3 Transform this ODE into a PDE.
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Definition (Random reinforced sequence)

Let (A_n) be a random sequence such that A_0 has law ν and

$$P(A_{j+1} = \ell | A_0, \dots, A_j) = (1 - q)\nu(\ell) + \frac{q}{j+1} \sum_{i=0}^j 1_{\{A_i = \ell\}}.$$

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We have $E(Z_n) = E(\prod_{j=0}^{n-1} A_j)$.

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We can see (A_n) as the sequence of balls drawn in a Polya urn, with $\frac{1}{j+1} \sum_{i=0}^j \delta_{A_i}$ encoding the composition of the urn.



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The sequence (A_n) can be constructed as the sequence of types in the following multitype Yule process:

- Each particle possesses a type $i \in \{0, \dots, k_\star\}$;
- A particle of type i gives birth to a particle of type i at rate $q + (1 - q)\nu(i)$, and to a particle of type j at rate $(1 - q)\nu(j)$.

Denote by $Y_t(j)$ the number of particles of type j at time t , and set

$$T_n = \inf \left\{ t > 0 : \sum_{i=0}^{k_\star} Y_t(i) = n \right\}.$$

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Lemma

Let $a = (a_0, \dots, a_{k_*}) \in \mathbb{R}_+^{k_*+1}$, we denote by

$$M_\ell(t, a) = \mathbb{E}_\ell\left(\prod_{j=0}^{k_*} a_j^{Y_t(j)}\right), \quad \text{where } \mathbb{P}_\ell(Y_0(\ell) = 1) = 1.$$

We have $\partial_t M_\ell(t, a) = M_\ell(t, a)(qM_\ell(t, a) + \phi(t))$, where $\phi(t) = (1 - q) \sum \nu(j) M_j(t, a) - 1$.

Corollary

Set $G(t, s) = \sum_{j=0}^{\ell} \frac{\nu(j) M_j(t, a)}{1 - s M_j(t, a)} = \sum_{j=0}^{\ell} \nu(j) \sum_{n=1}^{\infty} s^{n-1} M_j(t, a)^n$. We have

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Observing that G solves a transport equation with growth, we look for functions σ, γ such that $\gamma_t G(\sigma_t, t)$ is a constant. They solve

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If $m_{\nu, q} < 1$, then $Z_n \rightarrow 0$ a.s. as $n \rightarrow \infty$.

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Theorem (Bertoin–M. '25+)

- If $qk_* \geq 1$ or

$$\sum_{j=0}^{k_*} \frac{(1-q)j\nu(j)}{1-qj} > 1$$

then $P(\forall n \in \mathbb{N}, Z_n > 0) > 0$.

- If moreover $m_{*,q} = ((1-q)\nu(k_*) + q)k_* > 1$, then

$$\lim_{n \rightarrow \infty} \frac{Z_n}{m_{\nu,q}^n} = \lim_{n \rightarrow \infty} \frac{Z_n^*}{m_{*,q}^n} \quad \text{a.s.}$$

where (Z_n^*) is the largest k_* -ary subtree of the reinforced Galton-Watson process.

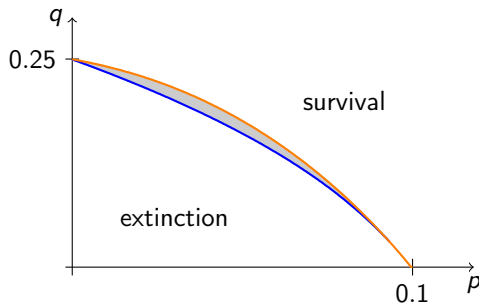


Figure: Phase diagram of a reinforced Galton-Watson process with parameter (ν_p, q) with $\nu_p = (1 - 4p)\delta_0 + p(\delta_1 + \delta_2 + \delta_3 + \delta_4)$, for $q \in [0, 0.25]$ and $p \in [0, 0.1]$. The blue line corresponds to (p, q) such that $m_{\nu_p, q} = 1$, the orange one such that $\sum \frac{(1-q)j\nu_p(j)}{(1-qj)} = 1$.



Lemma

The process $M_n = \sum_{|u|=n} \prod_{j<|u|} \frac{1}{m(L_{u_j})}$ is a positive martingale.

Proposition

By spine decomposition, this martingale is uniformly integrable if and only if

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Definition

We define the *pressure function* of the rGW(ν, q) as

$$\Lambda_q : \lambda \in \mathbb{R}^{k^*} \mapsto \log q - \log \left(\int_0^\infty \prod_{j=1}^{k^*} (1 - te^{\lambda(k)})_+^{\nu(k)(1-q)/q} dt \right).$$

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We have

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Theorem (Bertoin-M. 25+)

- ① With $\bar{\nu}_q := \nabla \Lambda_q(\ln)$, for all neighbourhood G of $\bar{\nu}_q$, there exists $\varepsilon > 0$ such that

$$E_q(\#\{|u| = n : L_u \notin G\}) \leq e^{-\varepsilon n} E_q(Z_n);$$

- ② Any law that satisfies $\langle \rho, \ln \rangle < \Lambda_q^*(\rho)$ is evanescent P_q -a.s.
- ③ Any law that satisfies $\langle \rho, \ln \rangle > H(\rho|q\rho + (1-q)\nu)$ is strongly persistent with positive probability.



Theorem (From Budhiraja and Waterbury ('22))

The sequence of empirical measures $(L_n = \frac{1}{n} \sum_{i=1}^n \delta_{A_k})$ satisfies the LDP under P_q with rate function $I_q : \mathcal{P}_S \rightarrow \mathbb{R}_+$ given by

$$I_q(\rho) = \inf_{\eta \in \mathcal{U}(\rho)} \int_0^1 H(\eta_s | q\psi_s + (1-q)\nu) ds,$$

where for all $t \in (0, 1]$

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Thank you for your attention!

